📄 VSEC PS-8 — FULL PROCESS (Ubuntu)

🔹 1. Install Required Software

Open Terminal and run:

sudo apt update
sudo apt install openjdk-17-jdk -y
sudo apt install git -y
sudo apt install python3 python3-pip -y

🔹 2. Verify Installation

java -version
git --version
python3 --version
pip3 --version

🔹 3. Download and Run Jenkins

Download .war file from Jenkins:

wget https://get.jenkins.io/war-stable/latest/jenkins.war

Run Jenkins:

java -jar jenkins.war

🔹 4. Unlock Jenkins

Open browser:

http://localhost:8080

Get password on another terminal:

cat ~/.jenkins/secrets/initialAdminPassword

Paste → Continue → Install suggested plugins → Create admin user

---

🔹 5. Create Project (Terminal)

```
mkdir AI-CI-Project
cd AI-CI-Project

touch app.py test_app.py requirements.txt Jenkinsfile
```

- 6. Add Code (Using nano)

```
nano app.py
```

Paste:

```
def add(a,b):
  return a+b
```

Save: CTRL + O → Enter → CTRL + X

```
nano test_app.py
```

Paste:

```
import unittest
from app import add

class TestAdd(unittest.TestCase):
  def test_add(self):
    self.assertEqual(add(2,3),5)

if __name__ == "__main__":
  unittest.main()
```

```
nano requirements.txt
```

Paste:

```
pytest
```

```
nano Jenkinsfile
```

Paste (Linux version):

```
pipeline {
  agent any

  stages {
    stage('Install Dependencies') {
      steps {
        sh 'pip3 install -r requirements.txt'
      }
    }

    stage('Run Tests') {
      steps {
        sh 'python3 -m unittest'
      }
    }
  }
}
```

- 7. Initialize Git

```
git init
git add .
git commit -m "Initial CI project"
```

- 8. Create Repo on GitHub

Go to GitHub

Create repo: AI-CI-Project

- 9. Connect GitHub

```
git remote add origin https://github.com/YOUR_USERNAME/AI-CI-Project.git
git branch -M main
git push -u origin main
```

🔹 10. Create Pipeline Job in Jenkins

Open Jenkins

👉 **New Item → Pipeline**

🔹 11. Configure Pipeline

Trigger:

✓ Select: Poll SCM

Add:

H/2 * * * *

Pipeline Section:

Field	Value
Definition	**Pipeline script from SCM**
SCM	**Git**
Repo URL	**Your GitHub repo: https://github.com/USER-NAME/AI-CI-Project.git**
Branch	***/main**
Script Path	**Jenkinsfile**

🔹 12. Run Build

Click:

Build Now

🔹 13. Check Output

Ran 1 test
OK
Finished: SUCCES